



Building the LoRa IoT platform

Part 2: update for PCB v5 (RAK3172)

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Wireless Sensors Made Simple
for agroecology & sustainable agriculture

Powered by technologies developed in  Intel-IrriS



List of components



For SOLAR



For 2xAA
alkaline
batteries



For SOLAR
with NiMH
battery

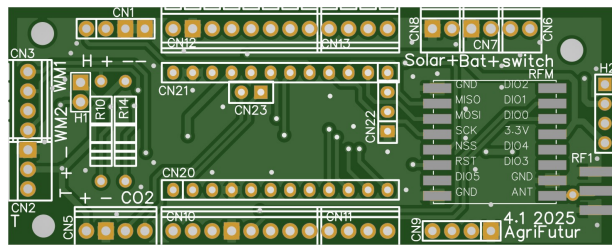


SMA male

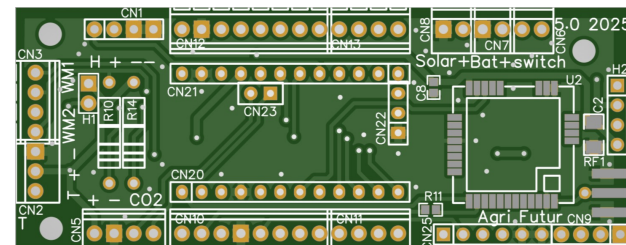


Important

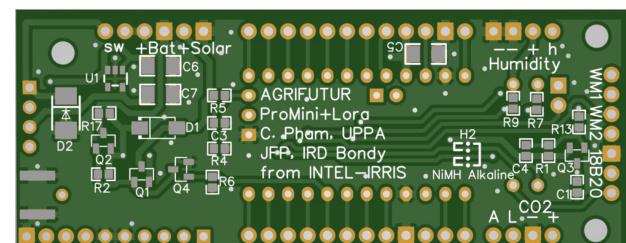
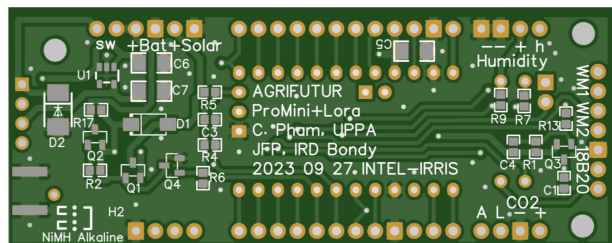
- ⦿ This tutorial is an update for the PCB v5 based on the RAK3172 radio module
- ⦿ Reader MUST first look at the tutorial presenting the PCB v4.1 to understand the common instructions
- ⦿ This tutorial only presents the differences between the 2 versions

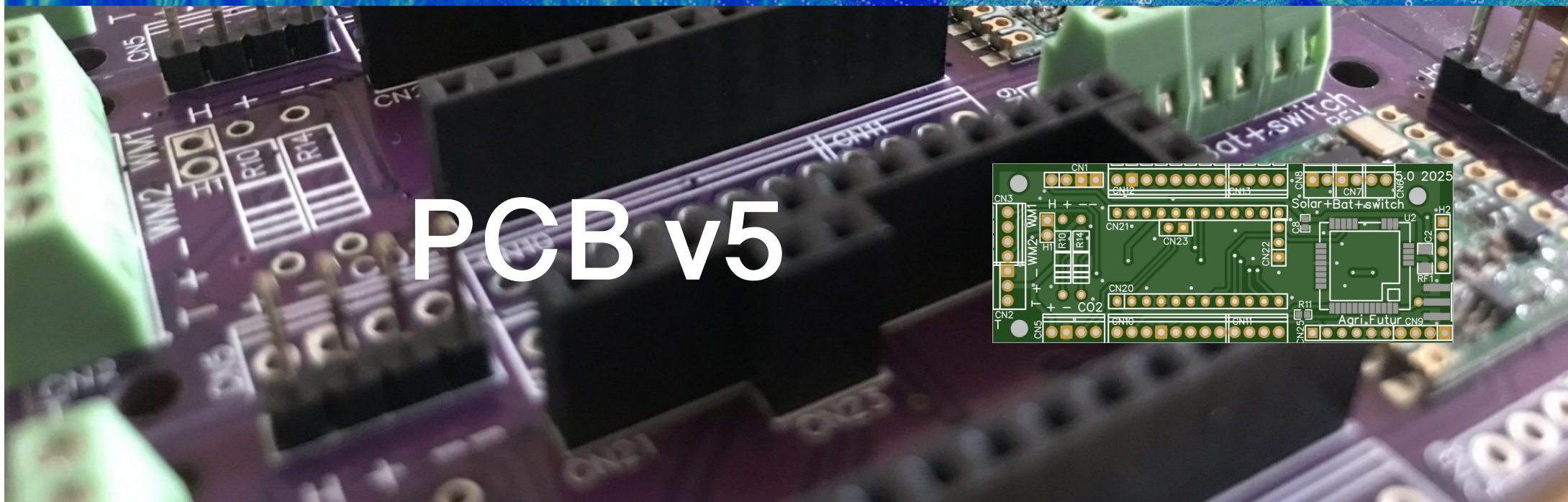


PCB v4.1
for
RFM95W
radio



PCB v5
for
RAK3172
radio

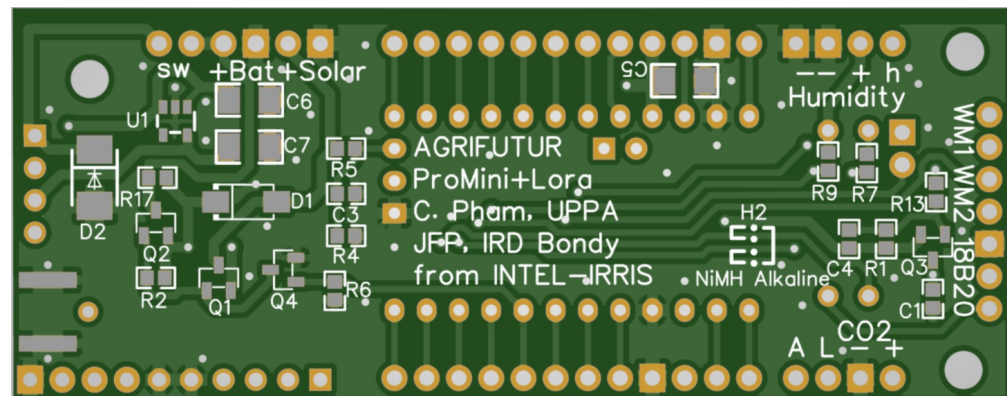
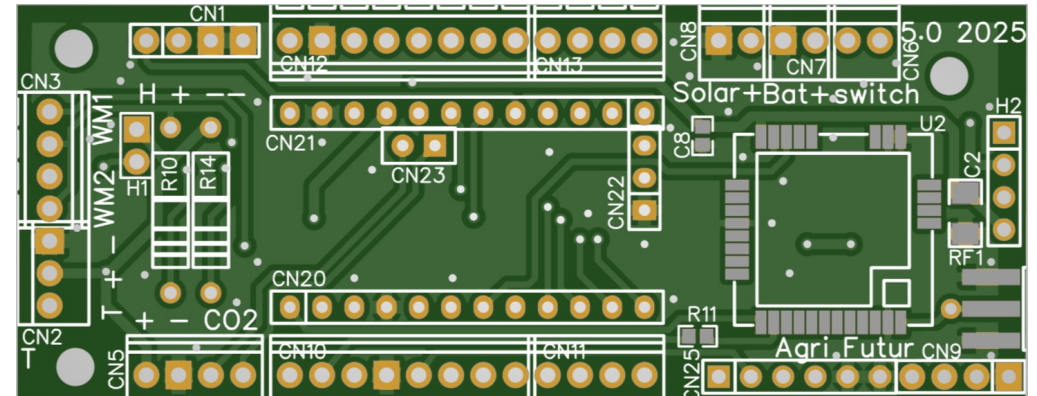






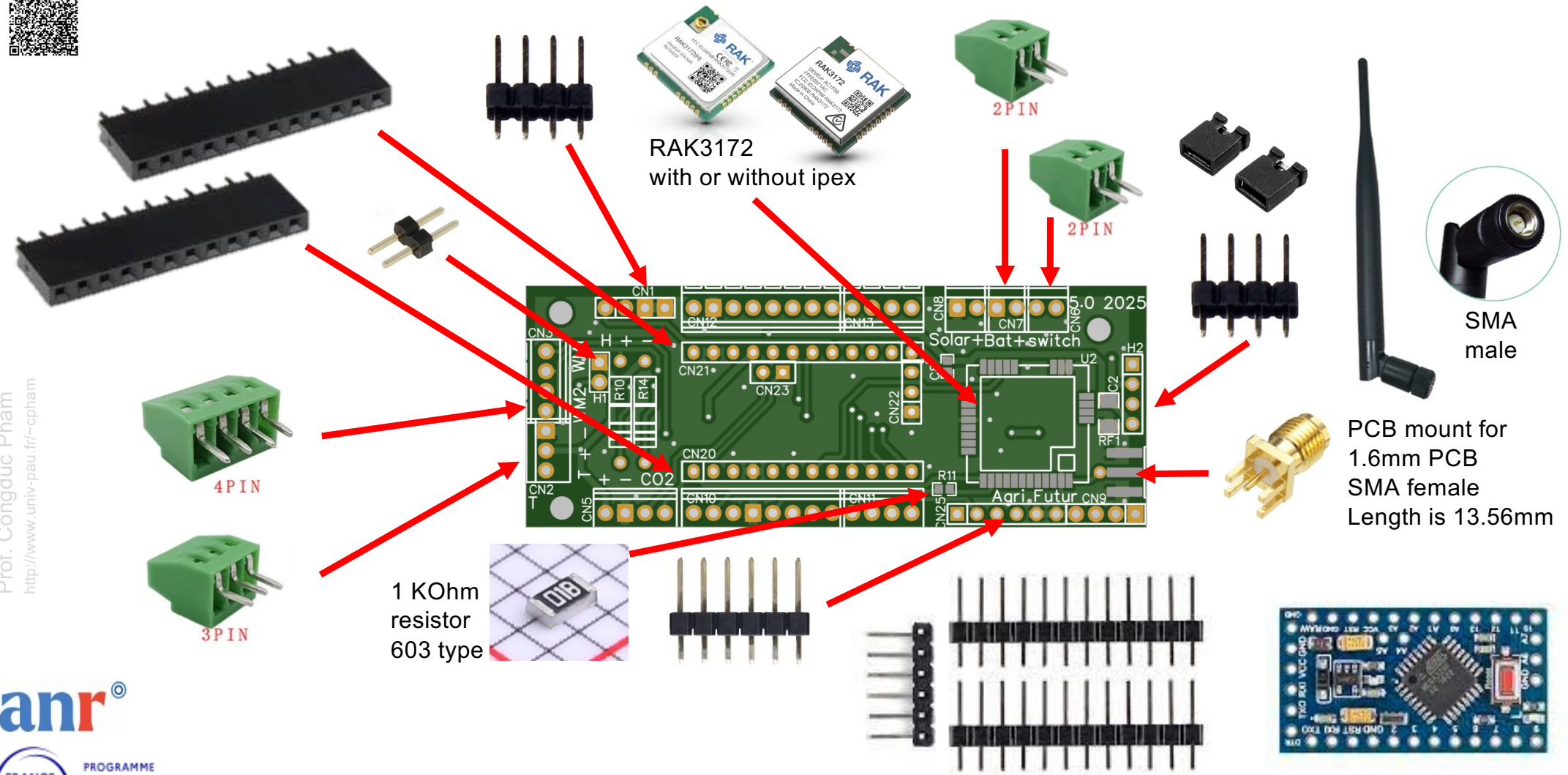
The PCB v5 – raw version

- You can order the raw version of the PCB, which means only the PCB, without any electronic components soldered by the manufacturer
- It is the so-called DIY approach
- The raw version is not intended to have solar charging capabilities





Electronic parts – starter-kit version

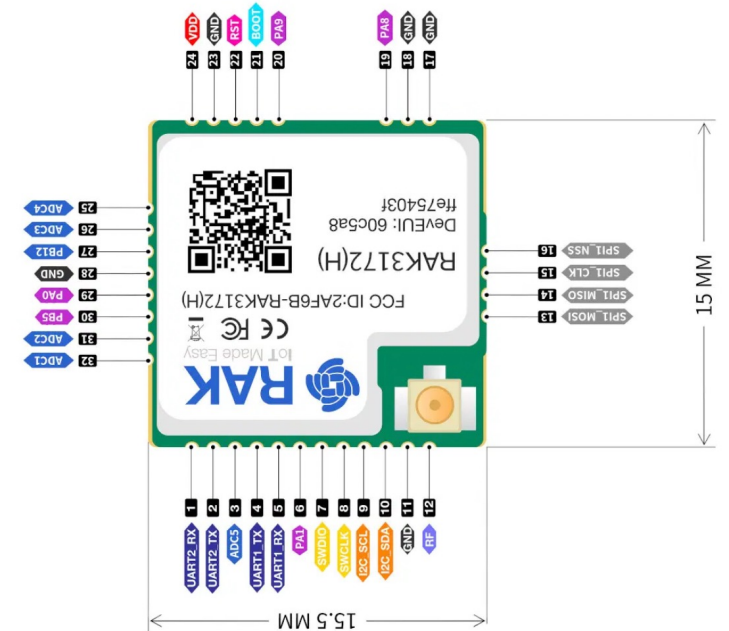
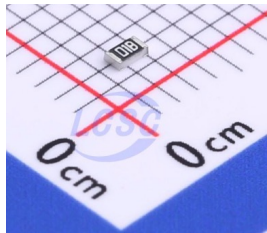
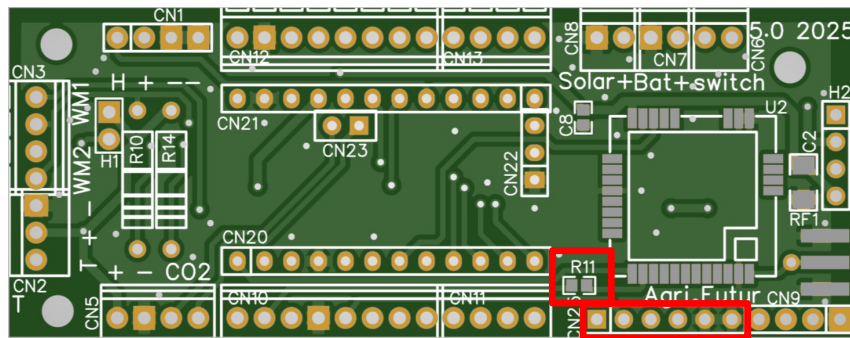


Arduino ProMini 3.3V 8MHz

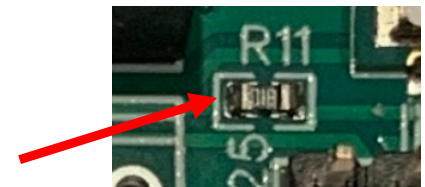


Difference compared to PCB v4.1

- ⦿ The RAK3172 radio replaces the RFM95W radio module. No need to solder all pins, only those marked by the red arrows



- ⦿ A 6-pin header needs to be soldered in order to enable serial monitor debug from the Arduino IDE
- ⦿ A 1KOhm resistor must be soldered at R11 place





PCBA v5



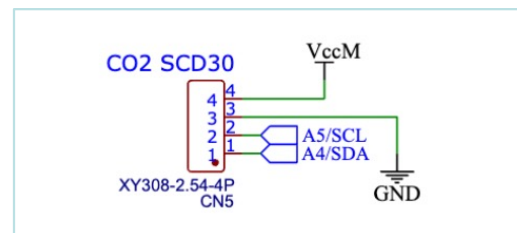
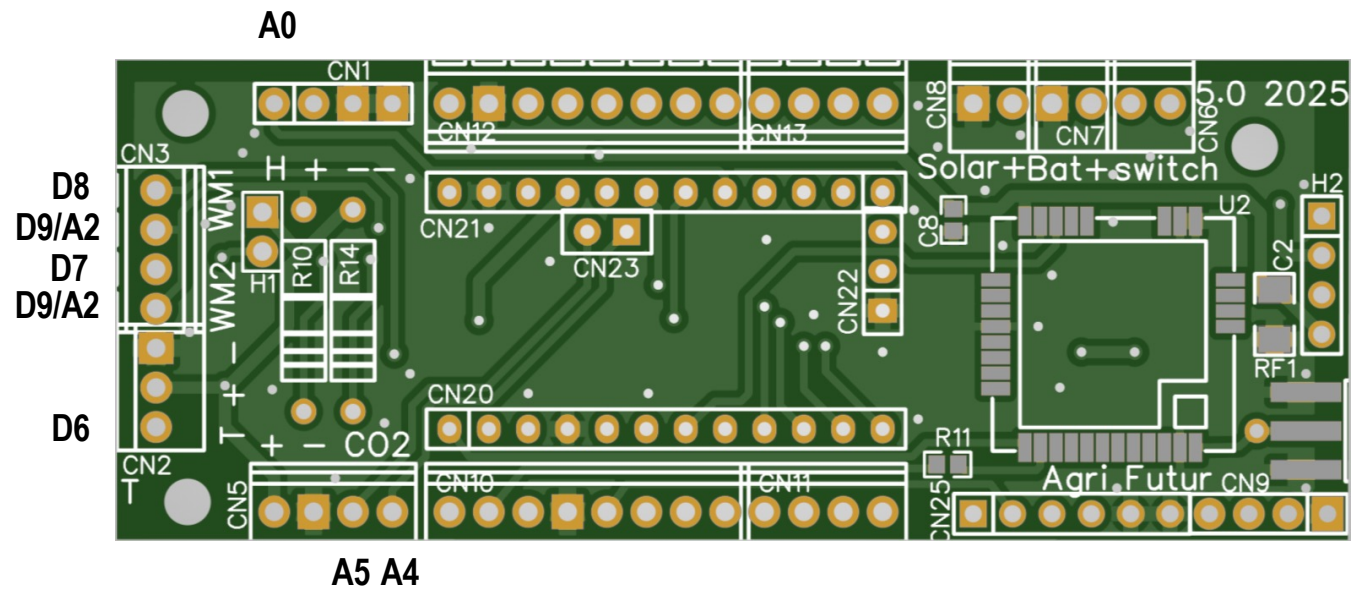
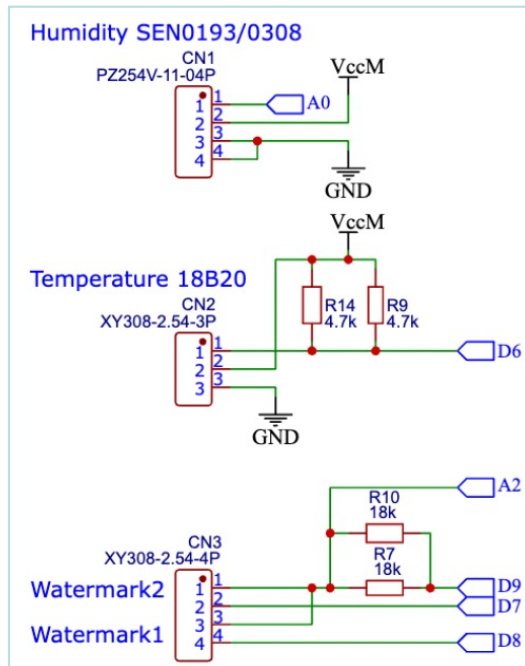
Image shown is the PCBA v5 from INTEL-IRRIS



PCB/PCBA v5 pin description

- The PCB has the following pre-defined wiring to Arduino ProMini pin

https://github.com/CongducPham/PEPR_AgriFutur/blob/main/PCBs/IRD_PCB_5/Schematic_IISS_5_2024-05-06.pdf





Soil device with PCBA v5, final result

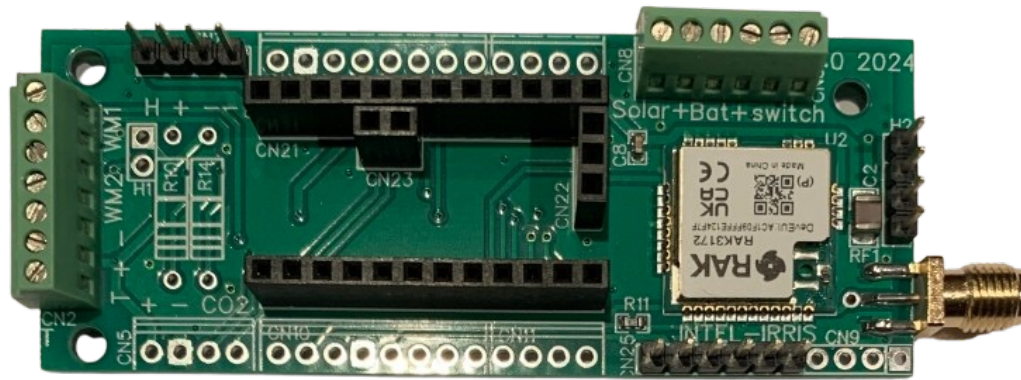
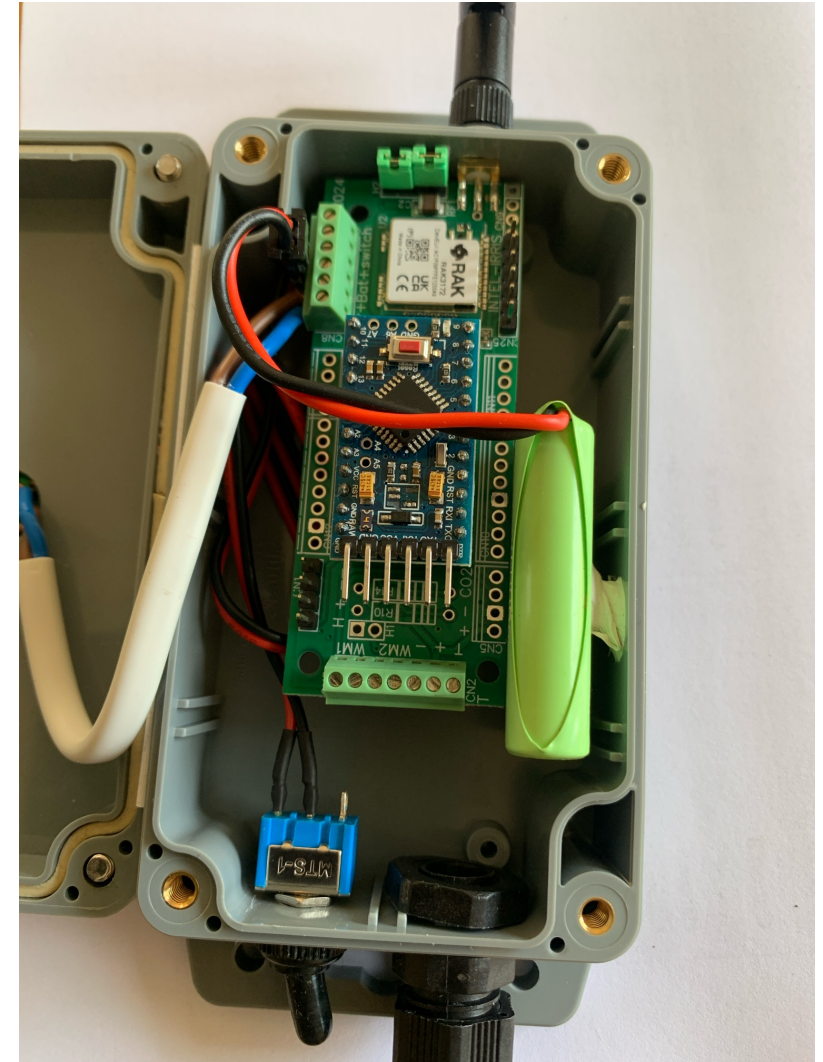


Image shown is the PCBA v5 from INTEL-IRRIS





PROGRAMME
DE RECHERCHE
AGROÉCOLOGIE
ET NUMÉRIQUE



PROGRAM THE
MICROCONTROLLER



Configuration for PCB/PCBA v5 (1)

- ⦿ For raw version, uncomment `IRD_PCB` in `BoardSettings.h`
- ⦿ For PCBA version, also uncomment `IRD_PCBA`

```

////////////////////////////////////
// uncomment for IRD PCB board (both v4.1 and v5)
#define IRD_PCB
// also uncomment for IRD PCB board fully assembled by manufacturer
// with all components, including solar circuit (both v4.1 and v5)
#define IRD_PCBA
    
```

- ⦿ **ONLY for solar version**, uncomment `SOLAR_BAT`

```

////////////////////////////////////
// uncomment only if the IRD PCB or PCBA is running on solar panel
// MUST be commented if running on alkaline battery
// code for SOLAR_BAT has been written by Jean-François Printanier from IRD
#define SOLAR_BAT
// do not change if you are not knowing what you are doing
#define NIMH
    
```




Config for PCB/PCBA v5 (2)

- ⦿ Uncomment `SOFT_SERIAL_DEBUG` in `BoardSettings.h`

```
////////////////////////////////////////
// uncomment for IRD PCB RAK version where
#define SOFT_SERIAL_DEBUG
```

- ⦿ Uncomment `RAK3172` in `RadioSettings.h`

```
////////////////////////////////////////
// please uncomment only 1 choice
// #define SX126X
// #define SX127X
// #define SX128X
#define RAK3172
////////////////////////////////////////
```



Serial monitor with PCB v5

- ⦿ After flashing the Arduino, it is usually desirable to watch at the text output from the Arduino IDE serial monitor to check that everything went OK
- ⦿ With PCB v5, the RAK radio module is controlled by AT commands received on its serial port and sent by the Arduino on its serial port (baud rate is normally set to 38400)
- ⦿ Therefore, Arduino's TX/RX pins are connected to the RAK3172 and the serial monitor used traditionally to look at text output can not use the default Arduino TX/RX pins
- ⦿ The PCB v5 uses a software defined serial port for the text output where only TX is defined on digital pin 2



Connect the FTDI USB-Serial

- After flashing the Arduino with the FTDI USB-Serial cable (left photo), unplug it in order to plug it again on the dedicated debug header on the PCB v5 (middle & right photo)

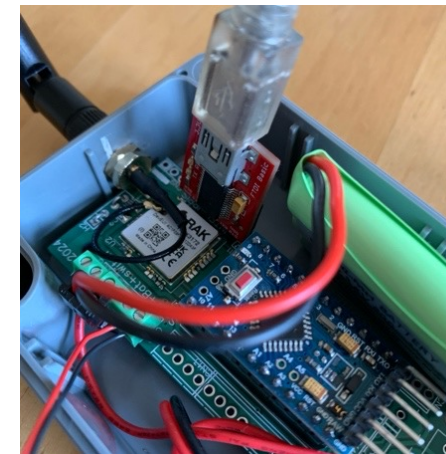
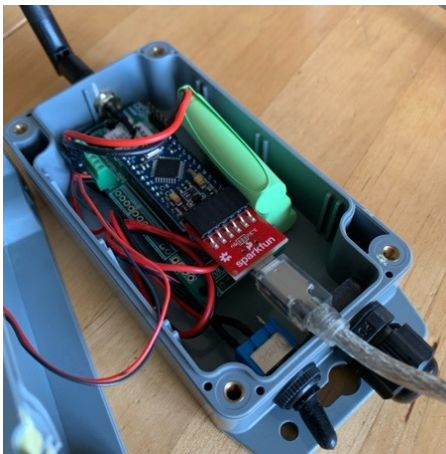
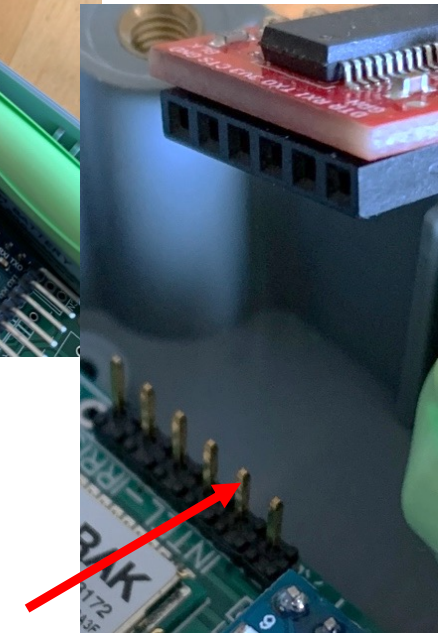


Image shown is the PCBA v5 from INTEL-IRRIS

- Battery should be connected, but switch is on OFF
- Be sure to connect the RX pin of the FTDI USB-Serial to the TX pin of the debug header on the PCB v5





Getting text output from the board

⦿ Order of steps is important!

```

/dev/cu.usbserial-AK05C87P

INTEL-IRRIS soil humidity sensor - Apr 29th, 2024
Arduino Pro Mini detected
ATmega328P detected
Opening serial port
Trying at 115200
-----
AT
Configuring for 38400
-----
AT+BAUD=38400
-----
AT
got answer
AT_COMMAND_NOT_FOUND
OK
ans.OK
-----
AT
got answer
OK
ans.OK
-----
AT+LPMVL=1
got answer
OK
  
```

Image shown is the Arduino IDE with INTEL-IRRIS code

Open serial monitor,
you should see nothing

Set baud rate to 38400

Switch ON the board

See output from board

Check that
transmission is OK



Annex: RAK3172 baud rate

- ⦿ Latest version of RAK3172 is using by default (factory setting) a **baud rate of 115200**, which is too fast (mostly for RX) for the Arduino Pro mini based on ATmega328P microcontroller
- ⦿ The device code is actually reprogramming the RAK3172 to use a baud rate of 38400, therefore you do not need to worry about setting the correct baud rate for RAK3172, especially with the fully assembled version of the PCB where the RAK3172 is provided and assembled on the PCB by the PCB manufacturer



Annex: IPEX or not IPEX?

- ⦿ The BOM file for the PCBA fully assembled is using the RAK3172 without the IPEX connector because we prefer to have a robust soldered RF connector

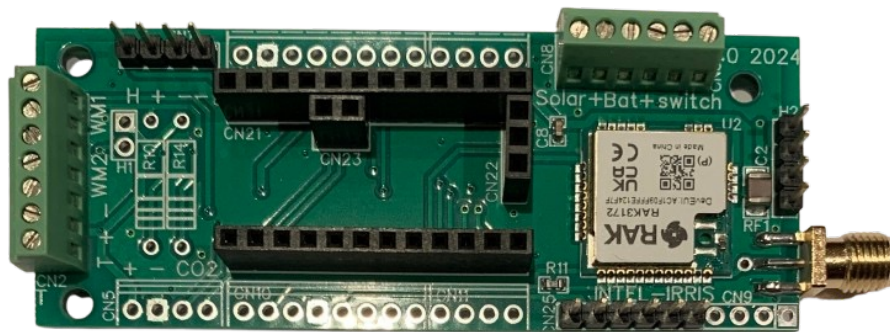
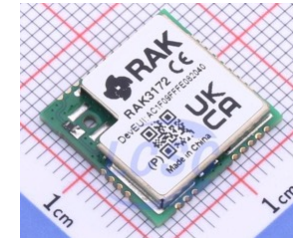
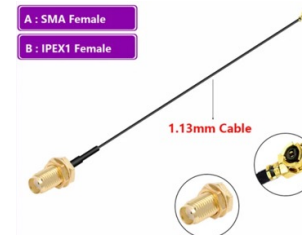
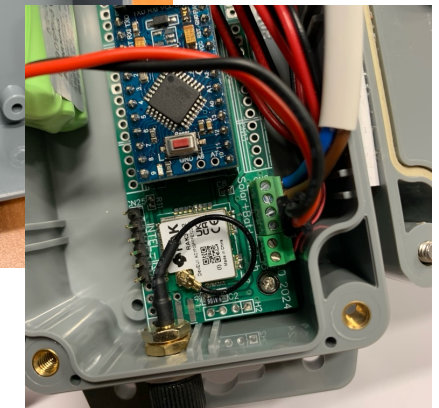


Image shown is the PCBA v5 from INTEL-IRRIS

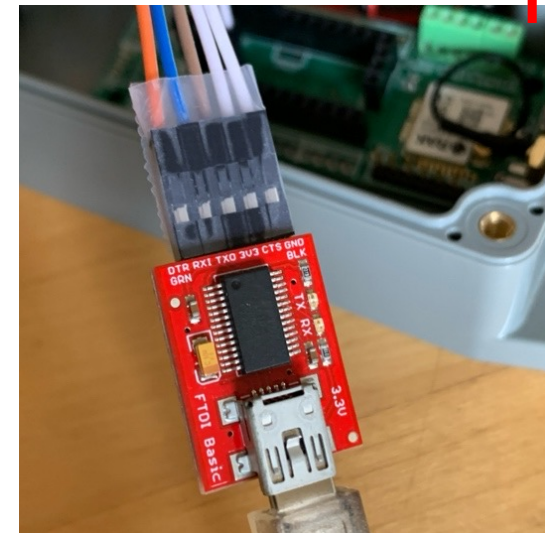
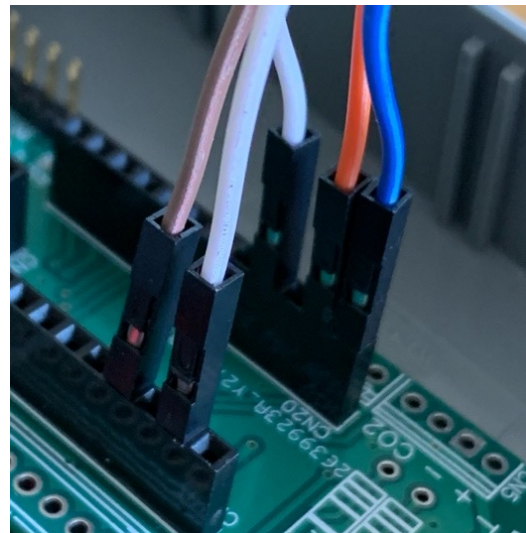
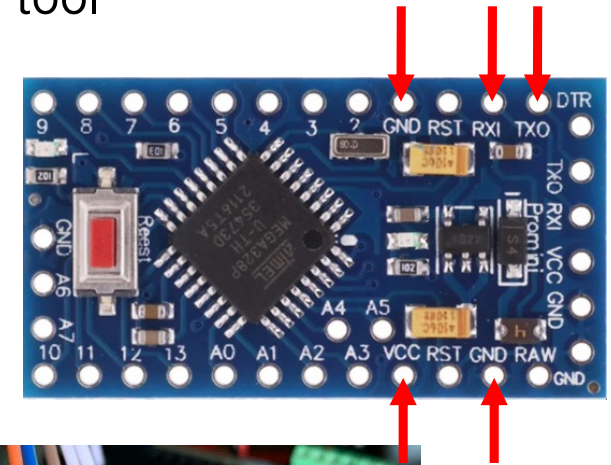
- ⦿ For the raw version of the PCB, you can solder the version with IPEX and connect the antenna with an IPEX-female to SMA female short pigtail





Annex: update firmware of RAK3172

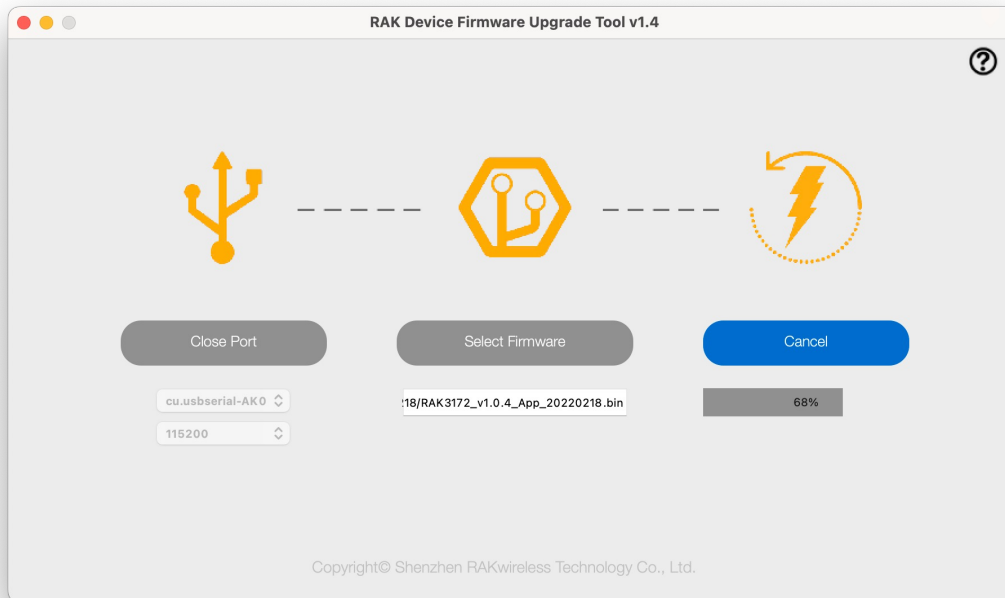
- The RAK3172 can be upgraded with the RAK DFU tool
- You can use Dupont wires to directly connect VCC, GND, RX and TX of the FTDI USB-serial cable to corresponding pins on the female headers after removing the Arduino Pro Mini from its socket. DTR is not connected





Annex: RAK DFU tool

- Get latest RAK3172 firmware from <https://downloads.rakwireless.com/#RUI/RUI3/Image/> usually, it is the RAK3172-E_latest.bin file
- Use RAK DFU tool to flash the new firmware



You may need to issue AT+BOOT from a serial tool in order to be able to upgrade with DFU tool



Annex: send direct AT commands

- With the FTDI USB-serial cable connected to the PCB v5 without the Arduino Pro Mini, as show on the previous slide, you can directly communicate with the RAK3172 to send AT commands for test purposes
- Any serial communication tool can be used, including the Arduino IDE serial monitor
- RAK also proposes the WisToolBox software to communicate with RAK components

```

AT+VER=?
AT+BAUD=38400
OK
AT+VER=RUI_4.0.6_RAK3172-E
OK
    
```

```

/dev/cu.usbserial-AK05C87P
AT+VER=?
AT+BAUD=38400
OK
AT+VER=RUI_4.0.6_RAK3172-E
OK
    
```

Autoscroll Show timestamp Both NL & CR 38400 baud Clear output



Annex: AT commands at startup

- ⦿ Here is the list of AT commands used by the device
- ⦿ During setup(), for ABP mode in EU868 band, for capacitive
 - ⦿ AT+BAUD=38400
 - ⦿ AT+LPMLVL=1
 - ⦿ AT+LPM=0
 - ⦿ AT+LPM=? ; AT+VER=? ; AT+BAND=? ; AT+ADR=? ; AT+NJM=? ; AT+DEVADDR=? ; AT+NWKSKEY=? ; AT+APPSKEY=?
 - ⦿ AT+NJM=0 ; AT+CLASS=A ; AT+BAND=4 ; AT+ADR=0
 - ⦿ AT+DEVADDR=26011DAA
 - ⦿ AT+NWKSKEY=23158D3BBC31E6AF670D195B5AED5525
 - ⦿ AT+APPSKEY=23158D3BBC31E6AF670D195B5AED5525
 - ⦿ AT+LPMLVL=2
 - ⦿ AT+LPM=1
- ⦿ On wake up, to transmit (sending 898.0 and 3.45 for instance)
 - ⦿ AT+SEND=1:0067231406020159
 - ⦿ AT+SLEEP

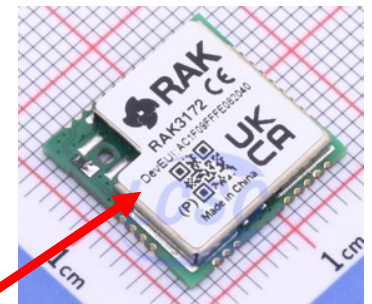


Annex: OTAA with RAK3172 (1)

- ⦿ The device with the RAK3172 can be integrated into a LoRaWAN ecosystem and configured by OTAA (Over The Air Activation) to dynamically get its address & session keys
- ⦿ Uncomment `ENABLE_OTAA` in `native_at_lorawan.h`

```
#include "BoardSettings.h"
```

```
#define ENABLE_OTAA
```



- ⦿ When using OTAA, you need to know
 - ⦿ DevEUI: unique identifier of the device, it is usually pre-provisioned by the radio module manufacturer and indicated on the radio module
 - ⦿ AppEUI: or JoinEUI, it is usually pre-provisioned by the radio module manufacturer. For RAK, it is usually AC1F09FFF8683172
 - ⦿ AppKey: it is usually pre-provisioned by the radio module manufacturer. For RAK it is usually DevEUI+AppEUI



Annex: OTAA with RAK3172 (2)

- ⦿ It is possible to use the default values pre-provisioned by the manufacturer for DevEUI, AppEUI and AppKey **IF YOU DIDN'T UPGRADE THE FIRMWARE**
 - ⦿ Leave `OVERWRITE_OTAA_EUI` commented
- ⦿ If RAK3172 firmware has been updated, they may not be set anymore! In this case, uncomment `OVERWRITE_OTAA_EUI` in `native_at_lorawan.cpp` and set in DevEUI, AppEUI and AppKey
- ⦿ Flash device and run once to set DevEUI, AppEUI and AppKey in the RAK module
- ⦿ Then, comment `OVERWRITE_OTAA_EUI` to flash again and run

```

////////////////////////////////////
//ENTER HERE your device info (same order, i.e. msb)
////////////////////////////////////
//define OVERWRITE_OTAA_EUI

#ifdef OVERWRITE_OTAA_EUI
//DevEui is normally indicated on the radio module
char* DevEuiStr="AC1F09FFFE12AAAA";
//here is the default used by RAK
char* AppEuiStr="AC1F09FFF8683172";
//here is the default used by RAK: DEVEUI+APPEUI
char* AppKeyStr="AC1F09FFFE12AAAAAC1F09FFF8683172";
#endif
    
```



Annex: register an OTAA device

- ⦿ To register your RAK3172 OTAA device on a LoraWAN Network Server such as TheThingNetwork (TTN) you can then just enter
 - ⦿ DevEUI: get it from the sticker
For instance **AC1F09FFFE12AAAA**
 - ⦿ AppEUI or JoinEUI: it is **AC1F09FFF8683172**
 - ⦿ AppKey: should be **AC1F09FFFE12AAAAAC1F09FFF8683172**
- ⦿ That's all. Once powered your device will join and you should receive data if a LoRaWAN gateway is near your device
- ⦿ **IMPORTANT**, the gateway is not full LoRaWAN in its single-channel version
- ⦿ We will discuss how to get data from an OTAA device, locally on the gateway in a dedicated tutorial

